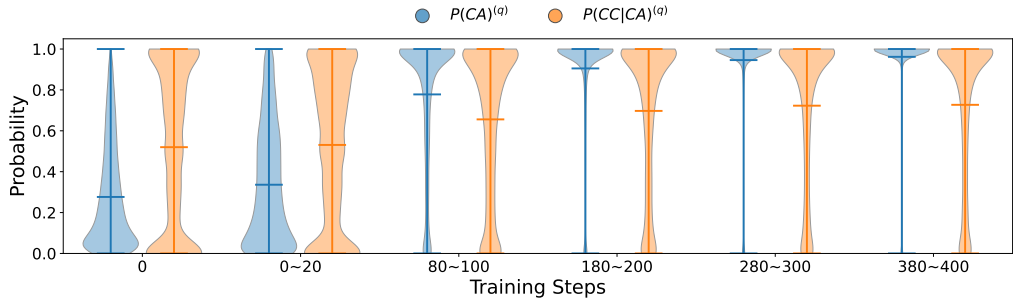


Figure 4: The evolution of $P(CA)^{(q)}$ (the fraction of correct answers for prompt q) and $P(CC|CA)^{(q)}$ (the fraction of correct CoTs within the correct answers for prompt q) for fully optimized training questions over the course of DAPO training.

Below we briefly illustrate our key perspectives on why GRPO works and when it may fail. Moreover, we include a complete proof for Theorem 1 and more discussions in Appendix A.5.

Discussions on the effectiveness of GRPO Theorem 1 indicates that even though a base LLM may guess the ground truth with imperfect CoTs at the beginning (low initial p_c^θ), GRPO could still work as long as the knowledge and logic priors have been established. The driving factor is the gap $\alpha - \beta > 0$, which amplifies the advantage difference between correct and incorrect CoTs. As training progresses and α increases (due to more sound reasoning across various question prompts) while β decreases (reducing spurious correlations, model biases, incorrect knowledge or calculation, etc.), causing the gap to widen and further accelerating coherent reasoning. As $p_c \to 1$, $(\alpha - \beta)$ may approach 1 in a faster pace because generating a few answer tokens is typically much easier than producing long correct CoTs, then $\mathbb{E}[\hat{A}(y_i) \mid \text{correct CoT}] \to 0$, ensuring convergence.

Discussions on failure modes in GRPO We note that the *Logic Prior* assumption may not always hold, potentially leading to the reinforcement of incorrect CoTs, since base LLMs may retain inherent biases and possibly fatal knowledge errors from pre-training. These harmful information might exist in some CoTs that finally yield the correct answer. In such cases, improper model biases could be unintentionally reinforced. We suspect that these unexpectedly reinforced CoTs are the root cause of the challenges faced by the R1-Zero approach (Guo et al., 2025), including poor readability and multi-lingual behaviors.

5 Training Dynamics of RLVR

To further demystify RLVR, we reproduce and analyze the training recipe of DAPO (Yu et al., 2025), which has been demonstrated to present extended reasoning capability boundaries in Section 3. Our experiments show that its training dynamics align pretty well with Theorem 1.

Key Indicators We first introduce key indicators that we have recorded during the reproduction. For each prompt q sampled with G responses, we define the number of answer passes and the number of both CoT and answer passes as $C = \sum_{i=1}^{G} \mathcal{I}_{\text{Ans}}(a_i)$ and $D = \sum_{i=1}^{G} \mathcal{I}_{\text{CoT}}(c_i) \cdot \mathcal{I}_{\text{Ans}}(a_i)$, respectively. We follow Chen et al. (2021)'s approach to calculate the Pass@K metric. Accordingly, we have per-prompt metrics: Pass@K$^{(q)} = 1 - \frac{\binom{G-C}{K}}{\binom{G}{K}}$ and CoT-Pass@K$^{(q)} = 1 - \frac{\binom{G-D}{K}}{\binom{G}{K}}$. Besides, we estimate the probability of producing correct answers for prompt q as $P(CA)^{(q)} = \frac{C}{G}$ and the probability of producing correct CoTs when generating correct answers as $P(CC|CA)^{(q)} = \frac{D}{C}$. For a dataset of multiple prompts, we take an average of per-prompt metrics as the dataset-level score.

Optimization Effects In our reproduced DAPO training, we observe that most training questions have been fully optimized. As shown in Figure 4, the probability of generating correct answers for these questions almost reach 1. In the meanwhile, we also observe the improvement in producing

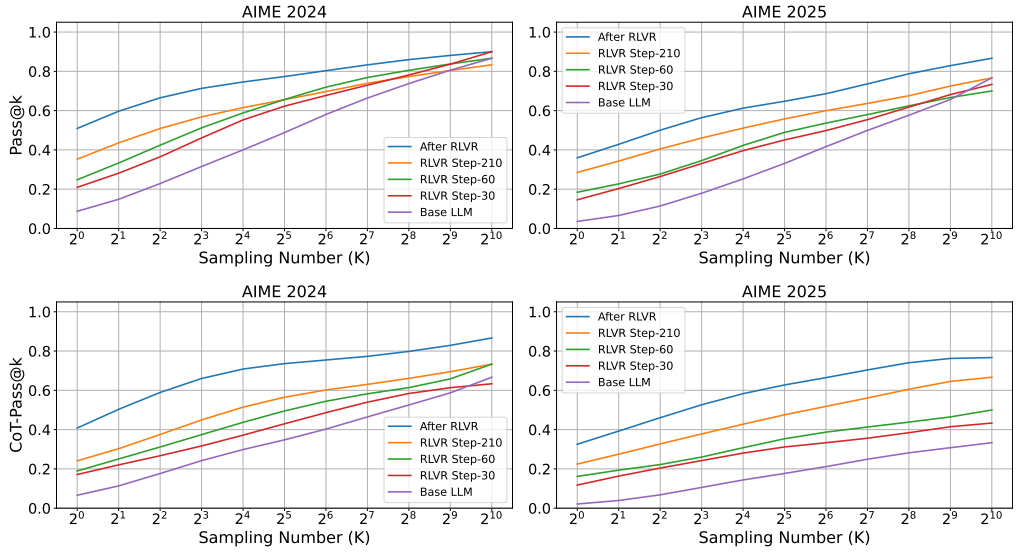


Figure 5: The evolution of Pass@K (the top row) and CoT-Pass@K (the bottom row) performance on AIME 2024 and 2025 for different model checkpoints during the DAPO training.

more correct reasoning CoTs, as indicated by the improvements in $P(CC|CA)^{(q)}$. These observations validate the key perspective in Theorem 1: RLVR not only optimizes the final verifiable reward but also implicitly incentivizes correct reasoning.

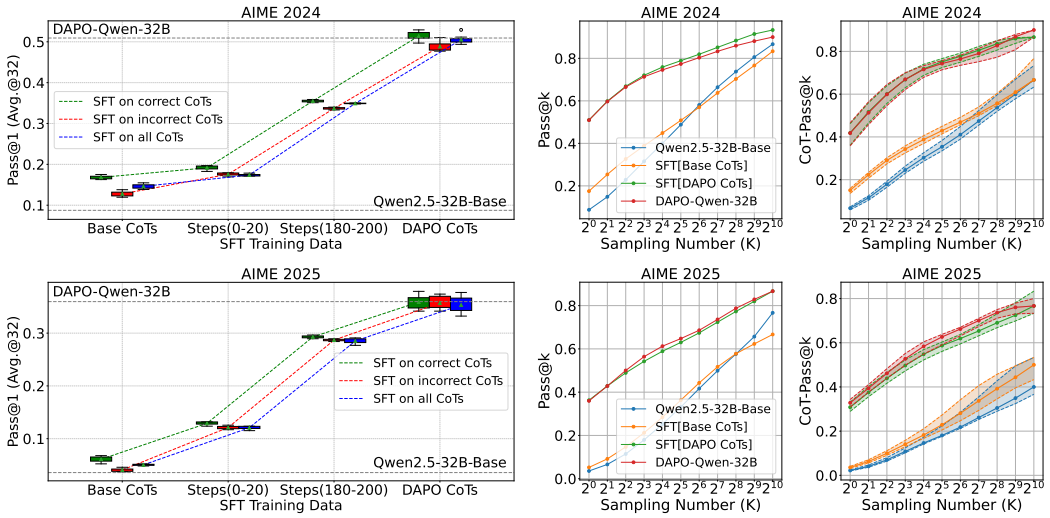
Generalization Behaviors In the meanwhile, Figure 5 discloses that the optimization of RLVR leads to the generalization improvements of both Pass@K and CoT-Pass@K from the very beginning. And using the CoT-Pass@K metric, we can clearly tell that the reasoning capability boundary has also been enhanced since the beginning. Another interpretation for this empirical observation is that the model has learned to produce more and more reasoning CoTs that DeepSeek-R1-0528-Qwen-8B cannot identify any error. This ability is learned implicitly since we have no formal supervision for the CoT correctness.

Limitations of DAPO Figure 4 also unveils the limitations of DAPO, the R1-zero approach applied to a 32B base LLM. As $P(CA)^{(q)}$ approaches 1.0 for most fully optimized training questions after 400 steps, which means these questions are no longer learnable because we cannot calculate a valid GRPO advantage for a all-correct group, we can still observe a non-negligible portion of imperfect CoTs (the median of $P(CC|CA)^{(q)}$ is around 0.7). These signals indicate that there are certain unexpected reasoning behaviors learned in DAPO and we may not have a chance to mitigate them purely based on answer correctness as the reward.

6 The Quality of Reasoning CoTs Enhanced by RLVR

In addition to the LLM-as-a-CoT-Judge approach for strictly identifying critical errors in reasoning CoTs, we further leverage supervised fine-tuning (SFT) to assess the quality of reasoning CoTs enhanced by RLVR. Given the training questions in DAPO, we conduct multiple SFT procedures, starting from the same base LLM and learning from CoTs generated by different models. If the CoT data is of high quality, we expect the post-SFT model to exhibit improved generalization performance. Figure 6 presents an overall quality evaluation of various CoT data.

Specifically, Figure 6(a) illustrates the evolution of CoT quality during RLVR. As training progresses, the generalization performance of post-SFT LLMs, measured in Pass@1, improves steadily. Ultimately, SFT on DAPO CoT data matches the Pass@1 performance of DAPO-Qwen-32B. This result indicates that, given sufficient training questions and CoT data from a post-RLVR model, we can replicate a new model with nearly the same Pass@1 performance simply through SFT. Moreover, an interesting observation is that, regardless of whether the CoT data contains identifiable errors, as



(a) The CoT quality at different RLVR stages, using Pass@1 on test sets as the proxy metric.

(b) The CoT quality before and after RLVR, using (CoT-)Pass@K on test sets as the proxy metric.

Figure 6: We show the generalization performance of post-SFT LLMs optimized on different CoT data. All these SFT processes start from the same base LLM, Qwen2.5-32B, with the only variable being the different CoT data on DAPO training questions. We use the performance on test sets (AIME 2024, 2025) as a proxy for the quality of the corresponding CoT data.

RLVR progresses, the CoT quality, measured in Pass@1, generally improves. This suggests that although some erroneous steps may be present, the overall quality of these "incorrect" CoTs in the later stages of RLVR improves significantly.

Figure 6(b) compares the CoT quality before and after RLVR, using both Pass@K and CoT-Pass@K as proxy metrics. Comparing DAPO-Qwen-32B with the post-SFT model trained on its CoT data, we observe that this simple SFT approach nearly replicates the performance of a post-RLVR model, which would otherwise require significant computational cost. When comparing Qwen2.5-32B with the post-SFT model trained on its CoT data, we find that the post-SFT model begins to mitigate guessing. This aligns with our expectations, as we only feed CoT data with correct answers from Qwen2.5-32B into the SFT procedure. We can therefore regard this process as a round of off-policy RLVR optimization. These observations indicate that the incentivized CoT data through RLVR is crucial, as such CoTs cannot be directly sampled from base LLMs. RLVR optimizes the model's reasoning abilities, ensuring that the generated CoTs are more accurate, coherent, and reliable, which is essential for handling complex tasks.

Limitations A key limitation of our study lies in the use of a LLM as the verifier for the correctness of reasoning CoTs, due to the prohibitive cost of manually checking a large volume of generated reasoning paths. Moreover, our theorem only explains the optimization process of RLVR but provides no guarantee for its generalization. We merely observe the generalization empirically. Due to space limitations, further discussions on the implications of our findings are deferred to Appendix A.7.

7 Conclusion

In this work, we address the fundamental problem of whether RLVR genuinely incentivizes novel reasoning in base LLMs. Through empirical evaluations and theoretical analysis, we justify a new perspective: RLVR implicitly incentivizes correct reasoning. Moreover, our analyses on training dynamics and CoT quality further confirm that the reasoning CoTs after RLVR are fundamentally different and can even help to replicate similar capabilities simply via supervised learning.

We hope these findings can not only resolve conflicting conclusions in prior work but also illuminate the untapped potential of RLVR in aligning LLMs with human reasoning systems. We envision a